

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR
INSTITUTE OF ENGINEERING AND TECHNOLOGY (IET), UDAIPUR
DCCN QUESTIONS

Data Communication and Computer Networks — B.Tech CSE

200 Viva Questions with Answers | 100 Core + 100 Tough / Conceptual

IET UDAIPUR

PART A — 100 CORE VIVA QUESTIONS

1. What is data communication?

Answer: Data communication is the exchange of data between two or more devices through a communication medium.

2. What are the five basic components of data communication?

Answer: Sender, receiver, message, transmission medium, and protocol.

3. What is a computer network?

Answer: A computer network is a group of interconnected devices that communicate and share resources.

4. What is a protocol?

Answer: A protocol is a set of rules that controls communication between network devices.

5. What is network topology?

Answer: Topology is the physical or logical arrangement of devices and links in a network.

6. Name common network topologies.

Answer: Bus, star, ring, mesh, tree, and hybrid.

7. What is LAN?

Answer: LAN is a Local Area Network covering a small geographic area such as a room, building, or campus.

8. What is MAN?

Answer: MAN is a Metropolitan Area Network covering a city or metropolitan area.

9. What is WAN?

Answer: WAN is a Wide Area Network covering large geographic areas and connecting networks over long distances.

10. What is the Internet?

Answer: The Internet is a global interconnection of networks that communicate using the TCP/IP protocol suite.

11. What is the OSI model?

Answer: OSI is a seven-layer reference model used to understand and standardize network communication functions.

12. Name the seven OSI layers.

Answer: Physical, Data Link, Network, Transport, Session, Presentation, and Application.

13. Which OSI layer transmits raw bits?

Answer: The Physical layer.

14. What is the function of the Data Link layer?

Answer: It provides node-to-node delivery, framing, error detection, and flow control.

15. What is the function of the Network layer?

Answer: It provides logical addressing and routing of packets between networks.

16. What is the function of the Transport layer?

Answer: It provides end-to-end delivery, segmentation, flow control, and reliability where required.

17. What is the Application layer?

Answer: It provides network services directly to applications, such as web, email, and file transfer services.

18. What is the TCP/IP model?

Answer: It is the protocol architecture used by the Internet, commonly described using Application, Transport, Internet, and Network Access layers.

19. Give one difference between OSI and TCP/IP.

Answer: OSI is a seven-layer reference model, while the commonly used TCP/IP model has four layers.

20. What is encapsulation?

Answer: Encapsulation is the process of adding protocol control information as data moves down the protocol stack.

21. What is decapsulation?

Answer: Decapsulation is the removal of protocol headers or trailers as data moves up the protocol stack at the receiver.

22. What is a signal?

Answer: A signal is an electrical, optical, or electromagnetic representation used to carry information.

23. What is an analog signal?

Answer: An analog signal varies continuously with time.

24. What is a digital signal?

Answer: A digital signal represents information using discrete levels, commonly binary 0 and 1.

25. What is frequency?

Answer: Frequency is the number of cycles of a periodic signal completed per second, measured in hertz.

26. What is amplitude?

Answer: Amplitude is the magnitude or strength of a signal.

27. What is phase?

Answer: Phase describes the position of a waveform relative to a reference.

28. What is bandwidth?

Answer: Bandwidth is the range of frequencies a communication channel can effectively carry.

29. What is data rate?

Answer: Data rate is the number of bits transmitted per second, measured in bps.

30. What is bit rate?

Answer: Bit rate is the number of bits transmitted per second.

31. What is baud rate?

Answer: Baud rate is the number of signal symbols transmitted per second.

32. What is transmission impairment?

Answer: Transmission impairment is degradation of a signal during transmission, such as attenuation, distortion, or noise.

33. What is attenuation?

Answer: Attenuation is the reduction in signal strength as it travels through a medium.

34. What is noise?

Answer: Noise is unwanted electrical or electromagnetic energy that interferes with a signal.

35. What is guided media?

Answer: Guided media carries signals through a physical path, such as twisted-pair cable, coaxial cable, or optical fiber.

36. What is unguided media?

Answer: Unguided media carries signals through free space, such as radio, microwave, or infrared.

37. What is twisted-pair cable?

Answer: It consists of pairs of insulated copper wires twisted together to reduce interference.

38. What is coaxial cable?

Answer: It is a cable with a central conductor, insulation, metallic shield, and outer protective layer.

39. What is optical fiber?

Answer: Optical fiber transmits data as light through a glass or plastic fiber.

40. Give two advantages of optical fiber.

Answer: It offers high bandwidth and low attenuation, and it is resistant to electromagnetic interference.

41. What is line coding?

Answer: Line coding converts digital data into a digital signal suitable for transmission over a medium.

42. What is Manchester encoding?

Answer: Manchester encoding represents each bit using a transition in the middle of the bit period.

43. What is modulation?

Answer: Modulation is the process of varying a carrier signal according to the information signal.

44. Name basic digital modulation techniques.

Answer: ASK, FSK, and PSK.

45. What is multiplexing?

Answer: Multiplexing combines multiple signals for transmission over a shared communication link.

46. What is TDM?

Answer: Time Division Multiplexing assigns different time slots to different signals.

47. What is FDM?

Answer: Frequency Division Multiplexing assigns different frequency bands to different signals.

48. What is switching?

Answer: Switching is the method of forwarding data through intermediate network devices toward its destination.

49. Name major switching techniques.

Answer: Circuit switching, packet switching, and message switching.

50. What is packet switching?

Answer: Packet switching divides data into packets that can be forwarded independently through a network.

51. What is framing?

Answer: Framing divides a data stream into manageable units called frames at the Data Link layer.

52. What is an error in data communication?

Answer: An error occurs when received data differs from the transmitted data.

53. Name common error-detection methods.

Answer: Parity check, checksum, and cyclic redundancy check (CRC).

54. What is parity checking?

Answer: Parity checking adds a parity bit to help detect certain bit errors.

55. What is CRC?

Answer: CRC is an error-detection technique based on polynomial division over binary data.

56. What is checksum?

Answer: Checksum is an error-detection value calculated from data blocks and checked at the receiver.

57. What is error correction?

Answer: Error correction detects errors and reconstructs the original data, either by retransmission or redundancy.

58. What is ARQ?

Answer: Automatic Repeat reQuest is a reliability technique in which erroneous or lost frames are retransmitted.

59. What is Stop-and-Wait ARQ?

Answer: The sender transmits one frame and waits for an acknowledgment before sending the next frame.

60. What is Go-Back-N ARQ?

Answer: If an error occurs, the sender retransmits the erroneous frame and subsequent outstanding frames.

61. What is Selective Repeat ARQ?

Answer: Only lost or erroneous frames are retransmitted.

62. What is sliding-window protocol?

Answer: It allows multiple frames to be outstanding before acknowledgments are required.

63. What is piggybacking?

Answer: Piggybacking combines an acknowledgment with an outgoing data frame in the reverse direction.

64. What is ALOHA?

Answer: ALOHA is a random-access method in which stations transmit when they have data to send.

65. What is Pure ALOHA?

Answer: In Pure ALOHA, a station transmits whenever it has data and retransmits after a collision.

66. What is Slotted ALOHA?

Answer: Slotted ALOHA restricts transmissions to synchronized time slots.

67. What is CSMA/CD?

Answer: Carrier Sense Multiple Access with Collision Detection senses the channel and detects collisions during transmission; it is associated with traditional shared Ethernet.

68. What is CSMA/CA?

Answer: Carrier Sense Multiple Access with Collision Avoidance attempts to reduce collisions, commonly used in wireless LANs.

69. What is an IP address?

Answer: An IP address is a logical address used to identify an interface on an IP network.

70. What is IPv4?

Answer: IPv4 is the fourth version of IP and uses 32-bit addresses.

71. What is IPv6?

Answer: IPv6 is the newer IP version and uses 128-bit addresses.

72. How many bits are in an IPv4 address?

Answer: 32 bits.

73. How many bits are in an IPv6 address?

Answer: 128 bits.

74. What is subnetting?

Answer: Subnetting divides an IP network into smaller logical networks called subnets.

75. What is a subnet mask?

Answer: A subnet mask identifies the network portion and host portion of an IPv4 address.

76. What is a default gateway?

Answer: A default gateway is the router used to reach destinations outside the local IP network.

77. What is routing?

Answer: Routing is the process of selecting paths and forwarding packets toward their destination.

78. What is a routing table?

Answer: A routing table contains destination information and the next-hop or interface used to forward packets.

79. What is unicast?

Answer: Unicast is one-to-one communication from one sender to one receiver.

80. What is broadcast?

Answer: Broadcast is one-to-all communication within the applicable broadcast domain.

81. What is multicast?

Answer: Multicast is one-to-many communication to a specific group of receivers.

82. What is congestion?

Answer: Congestion occurs when offered traffic exceeds the network's ability to handle it, causing delay or packet loss.

83. What is Quality of Service (QoS)?

Answer: QoS refers to mechanisms for managing network traffic to meet requirements such as delay, bandwidth, jitter, and loss.

84. What is the Transport layer's main purpose?

Answer: It provides process-to-process communication between applications on end systems.

85. What is UDP?

Answer: UDP is a connectionless transport protocol with low overhead and no built-in guarantee of delivery.

86. What is TCP?

Answer: TCP is a connection-oriented transport protocol that provides reliable, ordered byte-stream delivery.

87. What is the TCP three-way handshake?

Answer: It is the SYN, SYN-ACK, and ACK exchange used to establish a TCP connection.

88. What is a port number?

Answer: A port number identifies a communication endpoint associated with a process or service.

89. What is flow control?

Answer: Flow control prevents a fast sender from overwhelming a slower receiver.

90. What is the difference between TCP and UDP?

Answer: TCP provides connection-oriented reliable delivery; UDP is connectionless and has lower overhead without TCP's reliability mechanisms.

91. What is the Leaky Bucket algorithm?

Answer: It regulates outgoing traffic at a relatively fixed rate and can smooth bursts.

92. What is the Token Bucket algorithm?

Answer: It controls traffic using tokens that accumulate at a configured rate; tokens permit transmission and can allow controlled bursts.

93. What is DNS?

Answer: DNS is the Domain Name System that maps domain names to IP addresses and other DNS records.

94. What is HTTP?

Answer: HTTP is an application-layer protocol used for transferring web resources between clients and servers.

95. What is HTTPS?

Answer: HTTPS is HTTP protected using TLS to provide encrypted and authenticated communication.

96. What is FTP?

Answer: FTP is an application-layer protocol designed for transferring files between a client and server.

97. What is SMTP?

Answer: SMTP is an application-layer protocol used for sending and relaying email.

98. What is WWW?

Answer: The World Wide Web is a system of interlinked resources accessed over the Internet, commonly using HTTP/HTTPS.

99. What is network security?

Answer: Network security protects networked systems and data against unauthorized access, disruption, alteration, or disclosure.

100. Name the three classic security goals.

Answer: Confidentiality, integrity, and availability. Syllabus note: RTU's published DCCN syllabus covers introductory concepts, OSI/TCP-IP, physical layer, data-link error control and MAC protocols, IPv4/IPv6 and routing, transport protocols, QoS/bucket algorithms, and application-layer protocols/security. MLSU's B.Tech CSE scheme lists Data Communication and Computer Networks as BT4CS06/CT06. Exact unit ordering can vary by admission batch, so this PDF is intended as a consolidated viva revision set rather than an official university question bank.

PART B — 100 TOUGH / CONCEPTUAL VIVA QUESTIONS

101. Why is layering used in computer networks?

Answer: It divides complex communication into modular functions, standardizes interfaces, and limits the effect of changes between layers.

102. What is the difference between a service and a protocol?

Answer: A service defines what a layer offers to the layer above; a protocol defines how peer entities communicate to provide that service.

103. What is a service access point?

Answer: A logical interface through which one layer provides services to the layer above it.

104. Why is packet switching efficient for bursty traffic?

Answer: It uses statistical multiplexing, so link capacity is shared dynamically instead of being reserved for one connection.

105. What is statistical multiplexing?

Answer: Dynamic sharing of a link where transmission opportunities depend on current traffic demand.

106. What is store-and-forward switching?

Answer: An intermediate node receives and stores a packet before forwarding it.

107. What is propagation delay?

Answer: The time required for a signal to travel through the physical medium from sender to receiver.

108. What is transmission delay?

Answer: The time required to place all bits of a packet onto a link; it is packet size divided by link rate.

109. Differentiate propagation delay and transmission delay.

Answer: Propagation delay depends mainly on distance and propagation speed; transmission delay depends on packet size and link rate.

110. What is bandwidth-delay product?

Answer: The amount of data that can be in transit on a link, approximately bandwidth multiplied by propagation or round-trip delay as defined.

111. Why does higher bandwidth not necessarily reduce propagation delay?

Answer: Propagation delay is mainly determined by physical distance and propagation speed, not by the link's bit rate.

112. What is Nyquist's bit-rate formula for a noiseless channel?

Answer: For bandwidth B and L signal levels, the theoretical maximum is $2B \log_2(L)$ bits/s.

113. What is Shannon channel capacity?

Answer: For bandwidth B and signal-to-noise ratio S/N , the theoretical capacity is $B \log_2(1+S/N)$ bits/s.

114. Why are Nyquist and Shannon limits different?

Answer: Nyquist models a noiseless channel with finite signal levels, while Shannon incorporates channel noise.

115. What is signal-to-noise ratio?

Answer: The ratio of signal power to noise power, commonly expressed in decibels.

116. Why are decibels used in networking?

Answer: They express power or amplitude ratios logarithmically, making large gains and losses easier to represent.

117. What is baseband transmission?

Answer: Transmission in which the signal occupies the channel's base frequency range without first shifting it to a carrier band.

118. What is passband transmission?

Answer: Transmission in which information is carried by modulating a signal around a carrier frequency.

119. Why is modulation useful?

Answer: It allows information to be placed in a suitable frequency band, supports multiplexing, and makes practical wireless or band-limited transmission possible.

120. What is intersymbol interference?

Answer: Overlap between adjacent transmitted symbols at the receiver, causing decision errors.

121. What is equalization?

Answer: Signal processing used to compensate for channel distortion and reduce intersymbol interference.

122. What is the Nyquist zero-ISI criterion?

Answer: It specifies pulse-shaping conditions that permit ideal sampling without intersymbol interference.

123. What is a constellation diagram?

Answer: A signal-space plot showing modulation symbols, commonly using in-phase and quadrature components.

124. What is QAM?

Answer: Quadrature Amplitude Modulation encodes information by changing both amplitude and phase through I/Q components.

125. Why does higher-order modulation need better SNR?

Answer: Its symbols are closer together in signal space, so noise more easily causes an incorrect decision.

126. What is spread-spectrum transmission?

Answer: A technique that deliberately spreads a signal over a bandwidth larger than the minimum required.

127. Differentiate FHSS and DSSS.

Answer: FHSS changes among carrier frequencies according to a hopping pattern; DSSS multiplies data by a high-rate spreading code.

128. What is WDM?

Answer: Wavelength Division Multiplexing sends multiple optical channels using different wavelengths on the same fiber.

129. What is DWDM?

Answer: Dense WDM carries many closely spaced optical wavelengths over a single fiber.

130. Why is optical fiber immune to electromagnetic interference?

Answer: It carries information as light through a dielectric medium rather than as electrical current.

131. What is total internal reflection?

Answer: Complete reflection of light inside a medium when the incidence conditions exceed the critical angle.

132. What is modal dispersion?

Answer: Pulse spreading caused by different propagation paths in multimode fiber.

133. What is attenuation in optical fiber measured as?

Answer: Usually in decibels per kilometer.

134. What is bit stuffing?

Answer: Inserting an extra bit after a specified bit pattern so that a reserved frame flag cannot appear accidentally in payload data.

135. What is byte stuffing?

Answer: Inserting an escape byte before data bytes that would otherwise be interpreted as control characters.

136. Why is CRC stronger than a parity bit?

Answer: CRC can detect many multi-bit and burst errors that a single parity bit can miss.

137. What is Hamming distance?

Answer: The number of bit positions in which two codewords differ.

138. How many errors can a code with minimum distance d detect?

Answer: Up to $d-1$ bit errors.

139. How many errors can a code with minimum distance d correct?

Answer: Up to $\text{floor}((d-1)/2)$ errors.

140. What is forward error correction?

Answer: Adding redundancy so the receiver can correct certain errors without requesting retransmission.

141. Why can Stop-and-Wait have poor utilization on long-delay links?

Answer: The sender remains idle while waiting for an acknowledgment, even though the link may be capable of carrying more data.

142. How does a sliding window improve utilization?

Answer: It permits multiple frames to be in transit before acknowledgments arrive.

143. Why must Selective Repeat use careful sequence-number sizing?

Answer: Old delayed frames must not be mistaken for new frames after sequence numbers wrap around.

144. What is the key buffering difference between Go-Back-N and Selective Repeat?

Answer: Go-Back-N normally discards out-of-order frames; Selective Repeat buffers acceptable out-of-order frames.

145. Why is CSMA/CD unnecessary on full-duplex switched Ethernet?

Answer: There is no shared collision domain on a full-duplex point-to-point link.

146. Why does Wi-Fi use CSMA/CA instead of CSMA/CD?

Answer: A wireless station generally cannot reliably detect a collision while transmitting, so it tries to avoid collisions instead.

147. What is the hidden-terminal problem?

Answer: Two stations cannot hear each other but both can reach the same receiver, so their transmissions may collide.

148. How can RTS/CTS reduce hidden-terminal collisions?

Answer: It can reserve the wireless medium before the data transmission, reducing simultaneous transmissions from hidden stations.

149. What is the exposed-terminal problem?

Answer: A station unnecessarily defers because it senses a nearby transmission that would not actually interfere with its own receiver.

150. What is the purpose of the Ethernet FCS?

Answer: The Frame Check Sequence provides error detection, normally using CRC.

151. What is MTU?

Answer: Maximum Transmission Unit is the largest IP packet size a particular link can carry without requiring link-level fragmentation, subject to the protocol definition.

152. What is IPv4 fragmentation?

Answer: Splitting an IPv4 packet into smaller fragments so it can cross a link with a smaller MTU.

153. How is IPv6 fragmentation different?

Answer: IPv6 routers do not fragment packets; the source can use the Fragment extension header when needed.

154. What is CIDR?

Answer: Classless Inter-Domain Routing represents networks with prefix lengths instead of fixed address classes.

155. Why was CIDR introduced?

Answer: To reduce address wastage and help control the growth of Internet routing tables.

156. What is route aggregation?

Answer: Combining multiple related prefixes into one summary route when the same forwarding treatment applies.

157. What is longest-prefix matching?

Answer: Selecting the routing-table entry with the most specific matching destination prefix.

158. What is ARP?

Answer: A protocol used to discover the link-layer address corresponding to an IPv4 address on a local network.

159. Why is ARP normally local to a LAN?

Answer: The ARP request identifies a local next-hop/link-layer destination and is not routed like an IP packet.

160. What is IPv6 Neighbor Discovery?

Answer: An ICMPv6-based mechanism for neighbor resolution, router discovery, and related local-link functions.

161. What is ICMP used for?

Answer: It carries Internet-layer control and diagnostic messages such as error reports and echo requests/replies.

162. Why does ping use ICMP?

Answer: The common ping utility sends ICMP Echo Request messages and checks for Echo Replies.

163. What is TTL in IPv4?

Answer: A field that is decremented by routers to limit packet lifetime and prevent indefinite looping.

164. What is Hop Limit in IPv6?

Answer: The IPv6 equivalent of TTL for limiting packet lifetime across routers.

165. What is distance-vector routing?

Answer: A routing approach in which routers exchange distance/cost information with neighbors.

166. What is link-state routing?

Answer: A routing approach in which routers distribute link-state information and calculate paths from a topology database.

167. What is split horizon?

Answer: A loop-prevention technique that avoids advertising a route back through the interface from which it was learned.

168. What is count-to-infinity?

Answer: A distance-vector routing problem in which incorrect route metrics can increase gradually after a failure.

169. What is Dijkstra's algorithm used for?

Answer: It computes shortest paths from a source and is used by link-state routing protocols.

170. What is Bellman-Ford used for?

Answer: It provides the mathematical basis for distance-vector shortest-path calculations.

171. What is an autonomous system?

Answer: A collection of networks and routers under a common administrative policy for Internet routing.

172. What is BGP used for?

Answer: Border Gateway Protocol exchanges reachability information between autonomous systems and supports routing policy.

173. Differentiate IGP and BGP's inter-domain role.

Answer: An IGP operates within an autonomous system; BGP is used for inter-domain routing between autonomous systems.

174. What is NAT?

Answer: Network Address Translation changes address and/or port information as traffic crosses a translation device.

175. Why is NAT not a complete firewall?

Answer: NAT translates addresses; firewalling is about enforcing traffic-security policy. They may be deployed together but are different functions.

176. What is tunneling?

Answer: Encapsulating one protocol packet inside another so it can cross an intermediate network.

177. What is TCP flow control?

Answer: A mechanism using receiver feedback, including the advertised window, to prevent sender overflow of the receiver buffer.

178. What is TCP congestion control?

Answer: A mechanism that adjusts sending behavior according to perceived network congestion.

179. Differentiate flow control and congestion control.

Answer: Flow control protects the receiver; congestion control protects the network from overload.

180. What is TCP slow start?

Answer: A congestion-control phase in which the congestion window initially grows rapidly, typically exponentially per RTT.

181. What is congestion avoidance in TCP?

Answer: A phase in which the congestion window grows more cautiously, classically approximately linearly under additive increase.

182. What is TCP retransmission timeout?

Answer: A timer value used to decide when an unacknowledged segment should be retransmitted.

183. Why are TCP acknowledgments cumulative?

Answer: One acknowledgment can confirm receipt of all data up to a particular sequence position, reducing acknowledgment overhead.

184. What does a TCP sequence number identify?

Answer: The position of bytes in the TCP byte stream.

185. What does a TCP acknowledgment number indicate?

Answer: The next sequence number expected, thereby acknowledging all preceding bytes received in order.

186. Why is TCP called a byte-stream protocol?

Answer: It provides an ordered continuous stream of bytes to the application rather than preserving message boundaries.

187. Why is UDP useful for some real-time applications?

Answer: It has low overhead and avoids TCP's retransmission and connection-management behavior, which can be useful when timely delivery matters.

188. What is recursive DNS resolution?

Answer: A resolver obtains the requested DNS answer on behalf of the client by querying other DNS servers as necessary.

189. What is DNS caching?

Answer: Temporarily storing DNS responses according to TTL to reduce lookup delay and DNS server load.

190. Why is HTTP called stateless?

Answer: Each request can be processed independently at the protocol level; application state is maintained separately using mechanisms such as cookies or sessions.

191. What is MIME?

Answer: Multipurpose Internet Mail Extensions define mechanisms for representing different content types and email attachments.

192. What is an SMTP relay?

Answer: A mail server that accepts and forwards email toward another mail server.

193. What is SNMP?

Answer: Simple Network Management Protocol is used to monitor and manage network devices and services.

194. What is an SNMP MIB?

Answer: A Management Information Base is a structured collection of managed objects and their identifiers.

195. What is an SNMP trap?

Answer: An unsolicited event notification sent by an SNMP agent to a management system.

196. What is a Layer-3 switch?

Answer: A switch that can also perform IP routing functions in addition to Layer-2 forwarding.

197. What is a VLAN?

Answer: A logical Layer-2 network that separates broadcast domains within switching infrastructure.

198. Why are VLANs useful?

Answer: They provide logical segmentation, improve organization, and can isolate traffic without separate physical switches.

199. What is an 802.1Q trunk?

Answer: A link carrying traffic for multiple VLANs using IEEE 802.1Q tags.

200. What is a broadcast domain?

Answer: The set of devices that can receive a particular Layer-2 broadcast.

Syllabus note: The set is a consolidated viva-revision resource based on DCCN / Computer Networks topics appearing in RTU and MLSU curricula. It is not an official university question bank.